



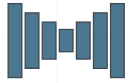
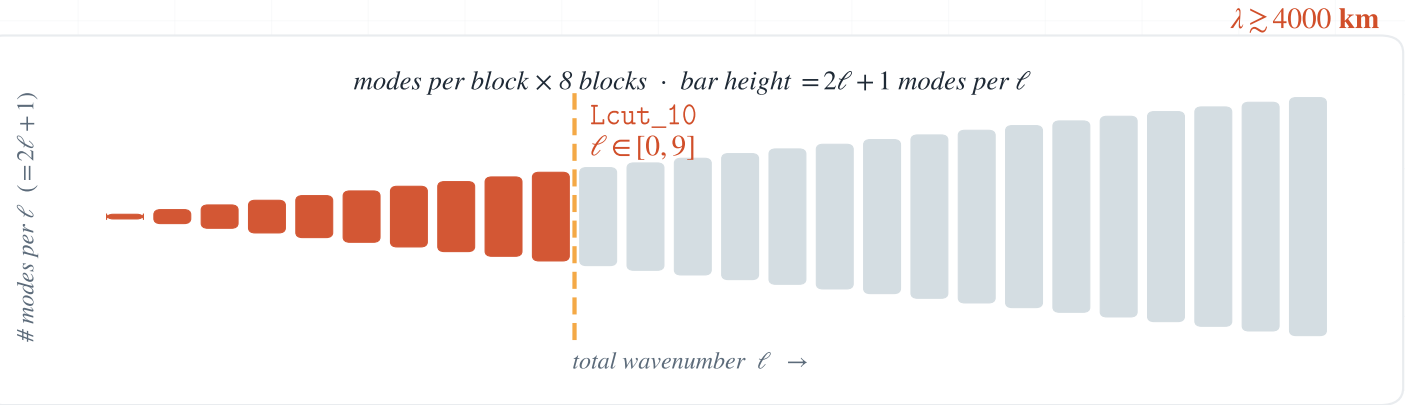
(a) SFNO

Spherical Fourier Neural Operator

Lcut_10

lowest ten modes $\ell \in [0, 9]$

spectral weight sub-slice



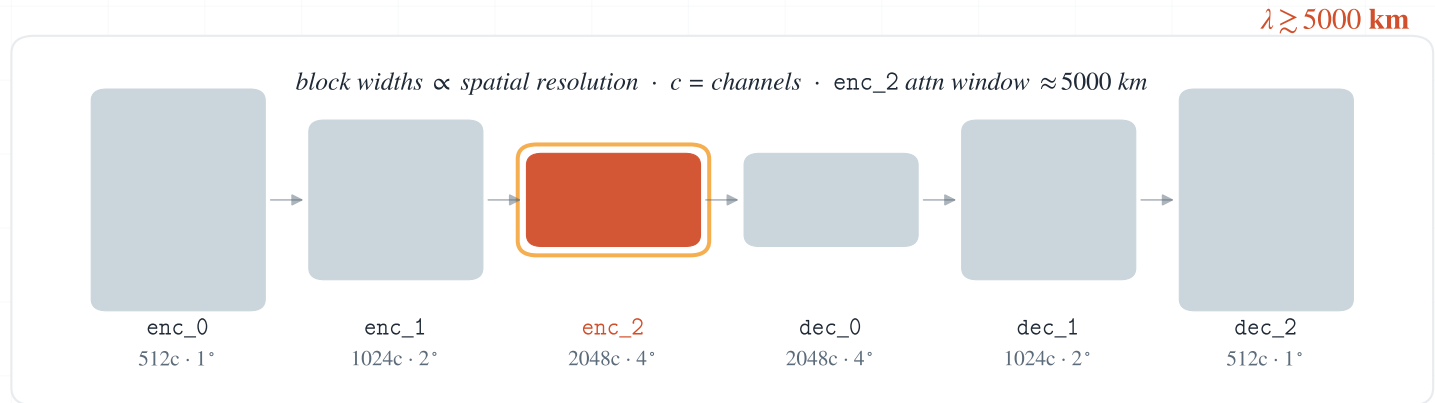
(b) Aurora

Swin-Transformer 3D U-Net

enc_2

encoder_layers.2 only

U-net encoder bottleneck



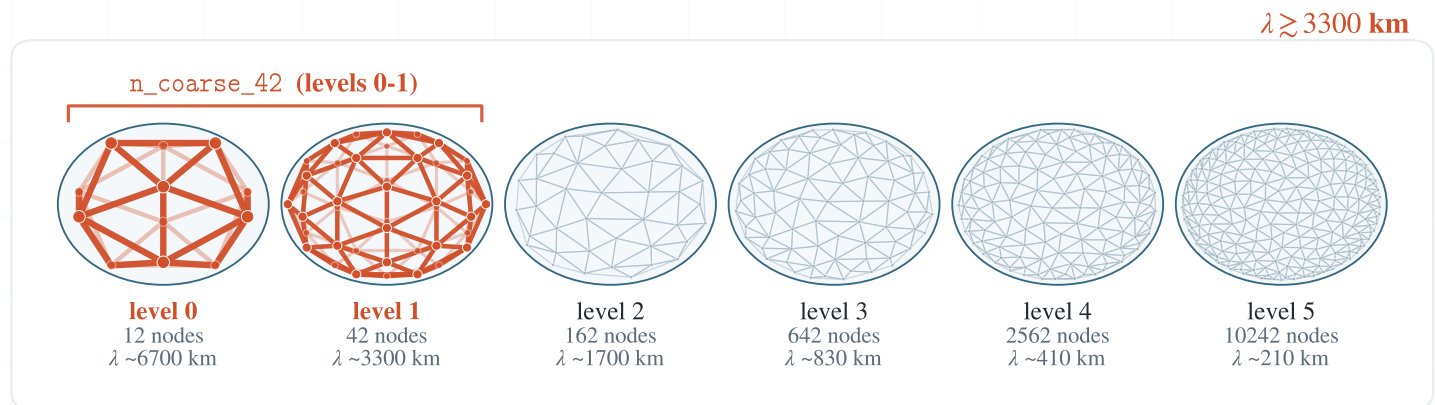
(c) GraphCast

Multi-Mesh Graph Neural Network

n_coarse_42

first 42 nodes (levels 0-1)

coarse-mesh activation hook



● targeted (perturbed)

● untouched (small-scale)